

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/DEB_SASL")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 3333, C2 = 6666, C3 = 1222)

text_priors <- '
Distrib (pAm,          TruncNormal_cv,          100,          0.2,          40, 2000);
Distrib (Fm,           TruncNormal,              0.1,           0.2,           0.01, 0.8);
Distrib (r_pAm_pM,     TruncNormal_cv,          0.8,           0.1,           0.1, 2);
Distrib (kap,          Uniform,                  0.1,           0.9);
Distrib (Ehp,          TruncNormal_cv,          100,           0.2,           40, 300);
Distrib (E_coc,        LogUniform,              10,            300);
Distrib (rOM_ClxFox,   LogUniform,              0.00001,       0.3);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIter)

# ./mcsim.DEBcaliS DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

	pAm.1.	Fm.1.	r_pAm_pM.1.	kap.1.	Ehp.1.	E_coc.1.
Result_mode	147.003000	0.08470200	1.04762000	0.60019700	143.37900	10.004200
2.5%	140.403000	0.07098710	1.00172000	0.56616900	121.55500	10.003400
97.5%	163.784000	0.13548400	1.15312000	0.61586700	170.81400	10.504900
Min	105.268000	0.05970440	0.81352300	0.29082400	75.44090	10.000000
Max	174.260000	0.44040700	1.21503000	0.85250600	191.28600	271.511000
Result_mean	151.396168	0.09518918	1.07267679	0.59208493	147.06205	10.150628
Result_sd	6.043837	0.01611522	0.03846049	0.01287943	12.27318	1.708858

	rOM_ClxHorse.1. Sigma_W.1.	
Result_mode	0.0099716700	0.14385000
2.5%	0.0031485075	0.12706510
97.5%	0.0131040000	0.16305660
Min	0.0000100002	0.11532500
Max	0.1516470000	4.24484000
Result_mean	0.0090769924	0.14385527
Result_sd	0.0024910353	0.03646418

Number of observations, n = 277
 Number of parameters, k = 7
 Log-likelihood, LL = -53.08438
 BIC = 145.5369

Potential scale reduction factors:

	Point est.	Upper C.I.
E_coc.1.	1.00	1.00
Ehp.1.	1.02	1.08
Fm.1.	1.02	1.09
kap.1.	1.03	1.10
pAm.1.	1.03	1.09
r_pAm_pM.1.	1.04	1.13
rOM_ClxHorse.1.	1.00	1.01
Sigma_W.1.	1.00	1.01

Multivariate psrf

1.03

```

mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)

```

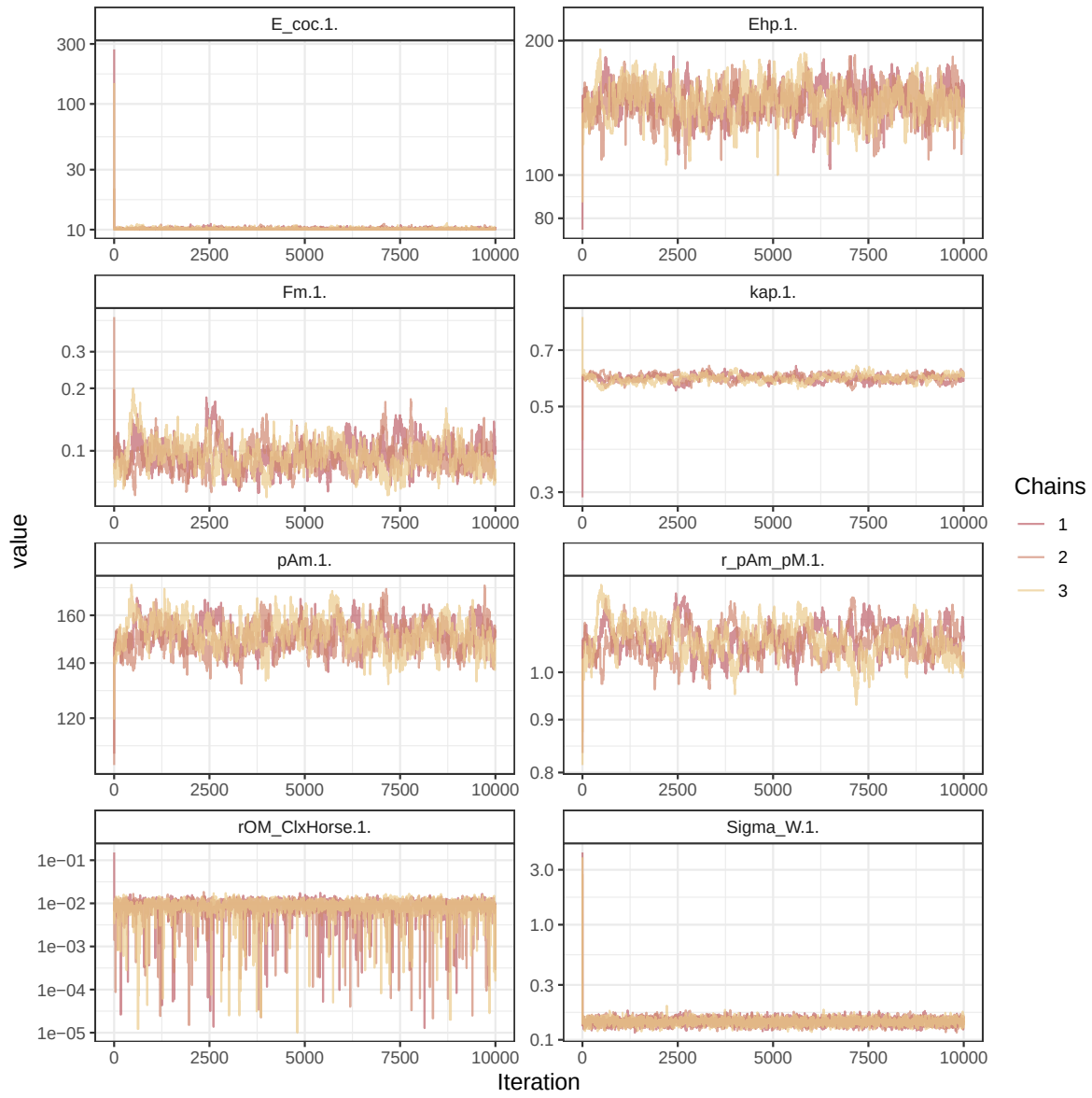


Figure 1: Parameters chains. Only the first iteration and the last `Nb_iter_kept` are represented.

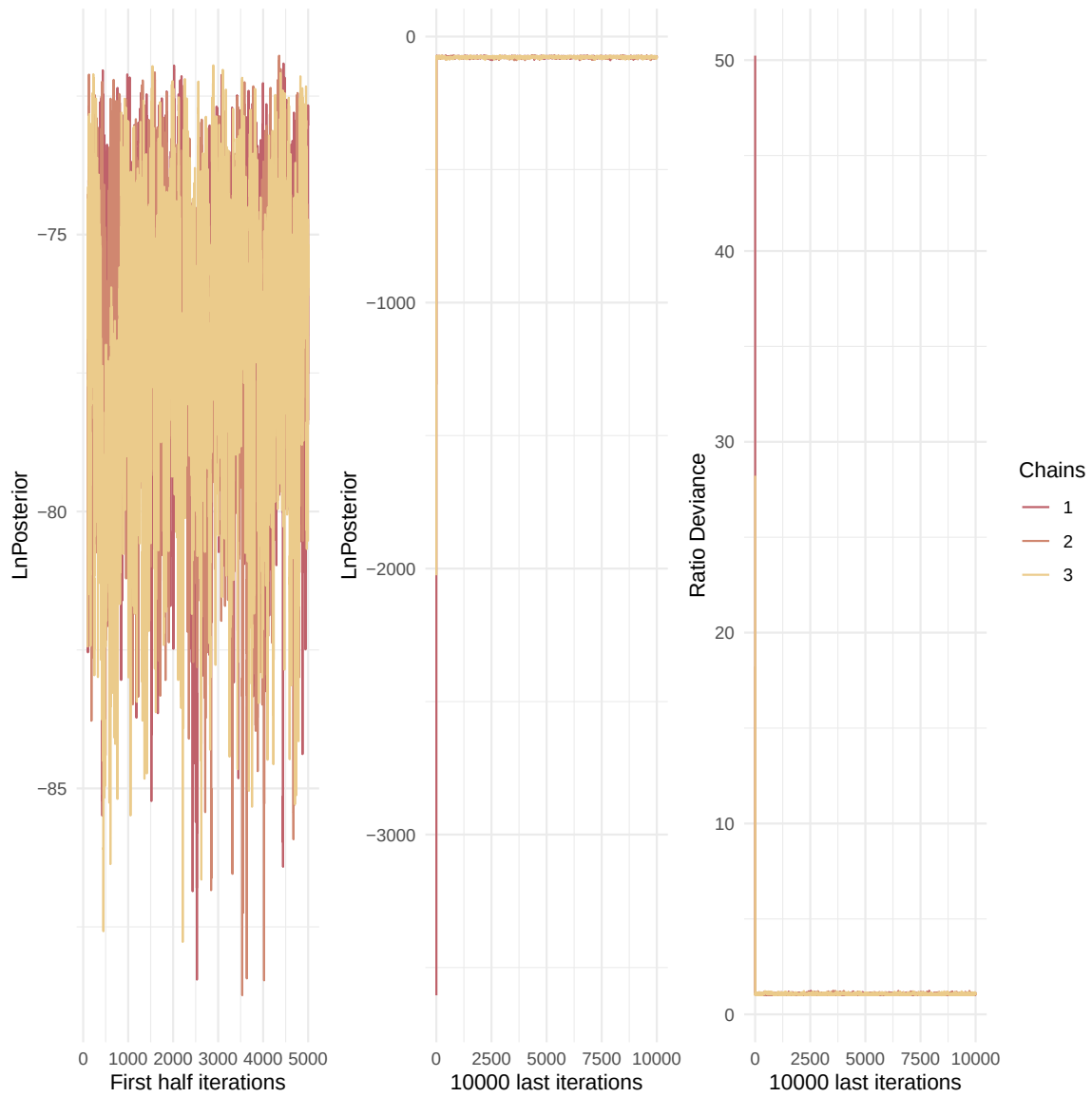


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

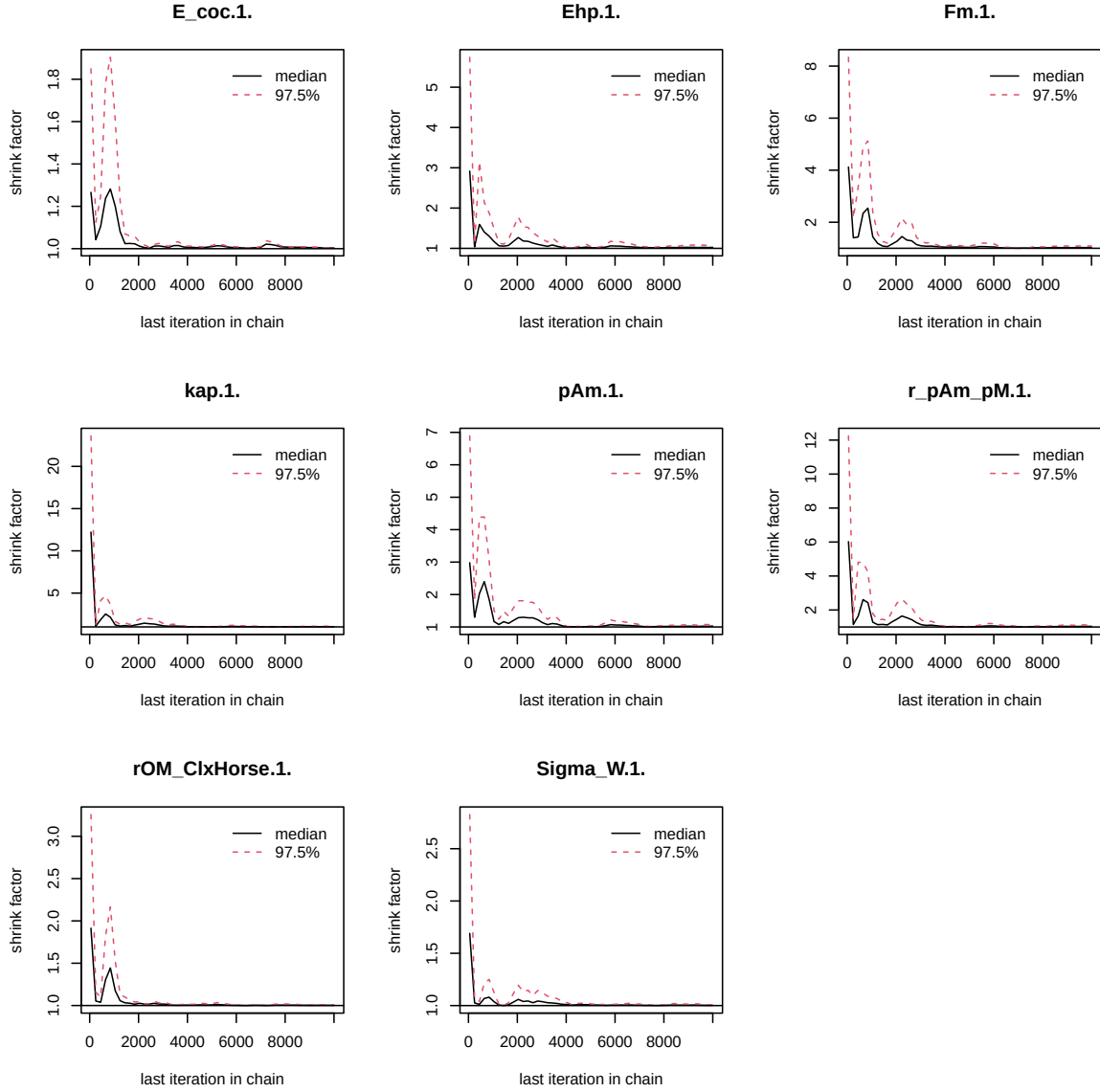


Figure 3: Chains convergence.

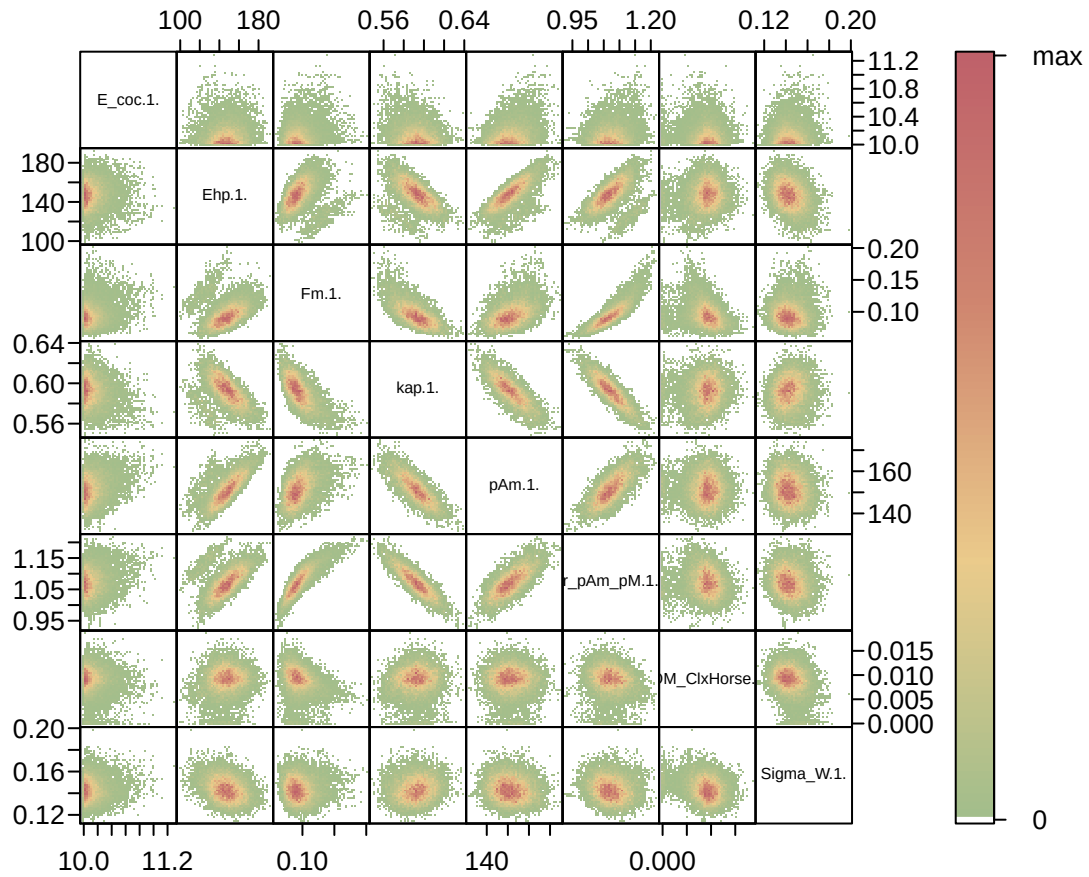


Figure 4: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

Posterior
and **prior** distributions

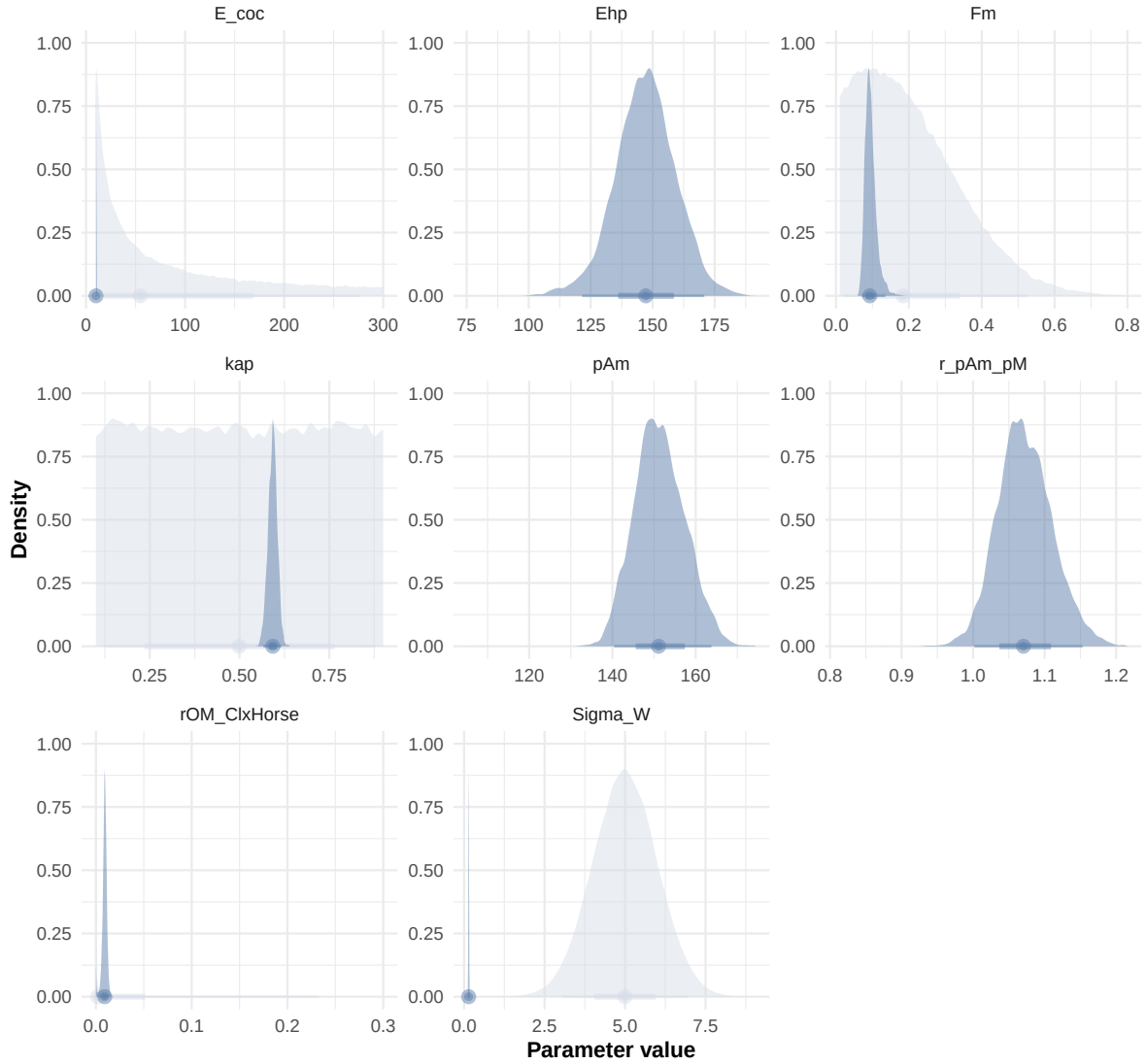


Figure 5: Distributions of the priors and the posteriors.


```
"Bart2020",  
#"Gollot2026_lufa"  
"Gollot2026_dens_D1",  
"Gollot2026_dens_D2",  
"Gollot2026_dens_D2b",  
"Gollot2026_dens_D5",  
"Gollot2026_dens_D7"  
)  
  
Adults_alone <- FALSE  
  
df_data <- f_import_data_DEB(l_Experiments, Adults_alone)  
  
text_param <- Summary_res["Result_mode", , drop = FALSE] %>%  
  t() %>%  
  as.data.frame() %>%  
  tibble::rownames_to_column("param") %>%  
  rename(val = 2) %>%  
  mutate(  
    param = gsub("\\\\.1\\.$|\\.1\\.\"", "", param),  
    line = glue("{param} = {val};")  
  ) %>%  
  pull(line) %>%  
  paste(collapse = "\n")  
  
Endpoints_print <- "Weight,  
  Reproduction"  
  
f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_print)  
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pr  
  
# ./mcsim.DEBcaliSL DEB_sim_full.in  
  
#####  
df_No_sim <- df_data |>  
  dplyr::select(ID_experiment, No_sim)  
  
df_carac_color <- data.frame(  
  carac = unique(l_carac),  
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],  
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])  
)
```

```

v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times)

```

Warning: `read\_table2()` was deprecated in readr 2.0.0.
 i Please use `read\_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpected relationship between variables
 i Row 1 of `x` matches multiple rows in `y`.
 i Row 1 of `y` matches multiple rows in `x`.
 i If a many-to-many relationship is expected, set `relationship = "many-to-many"` to silence this warning.

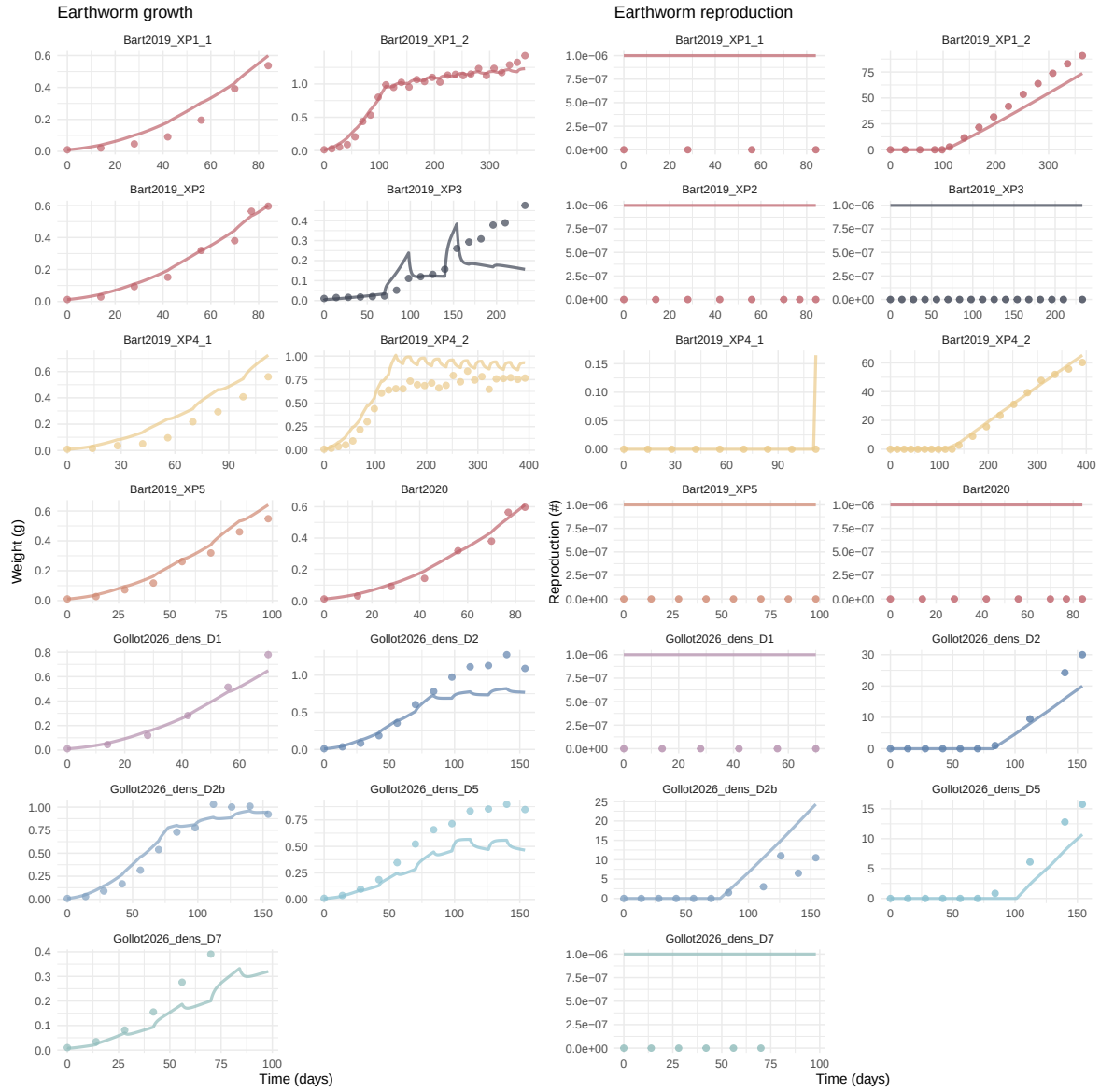
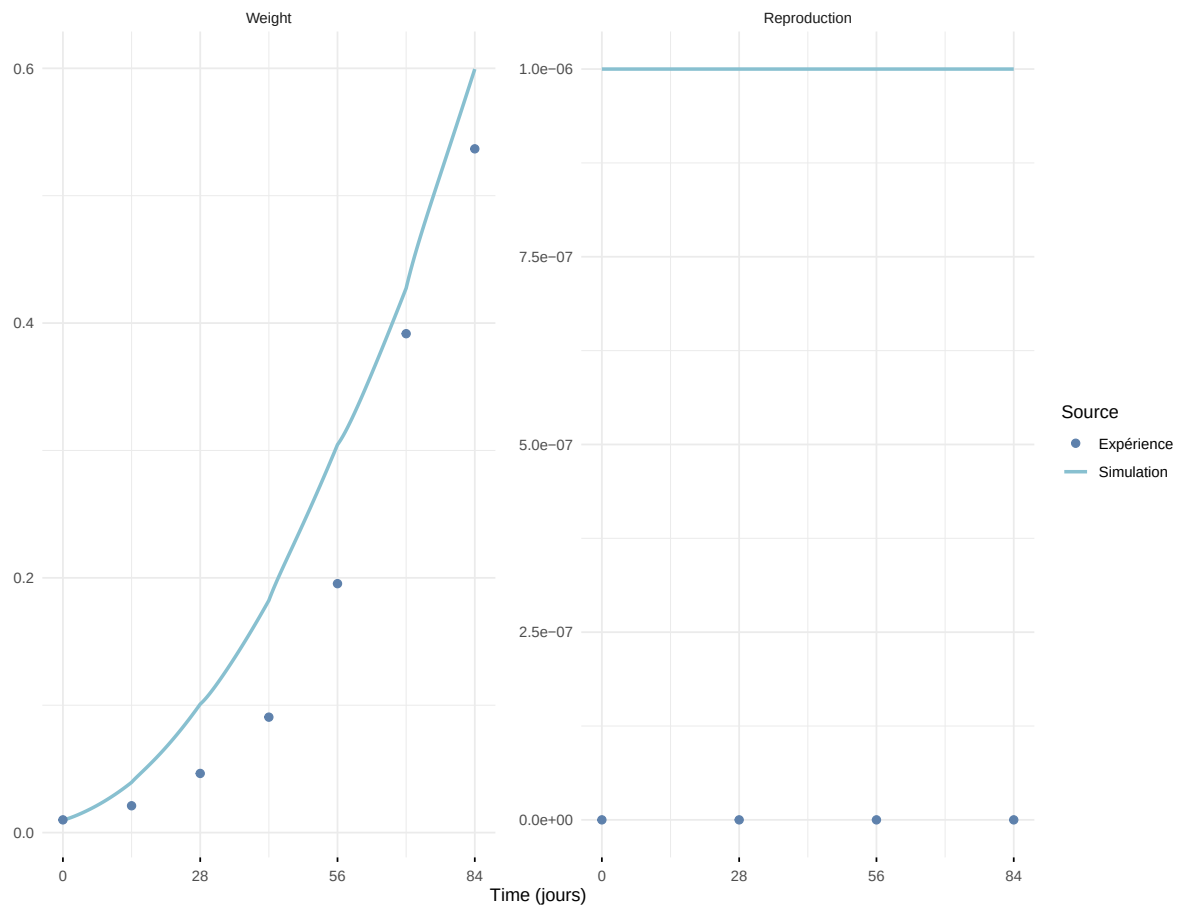
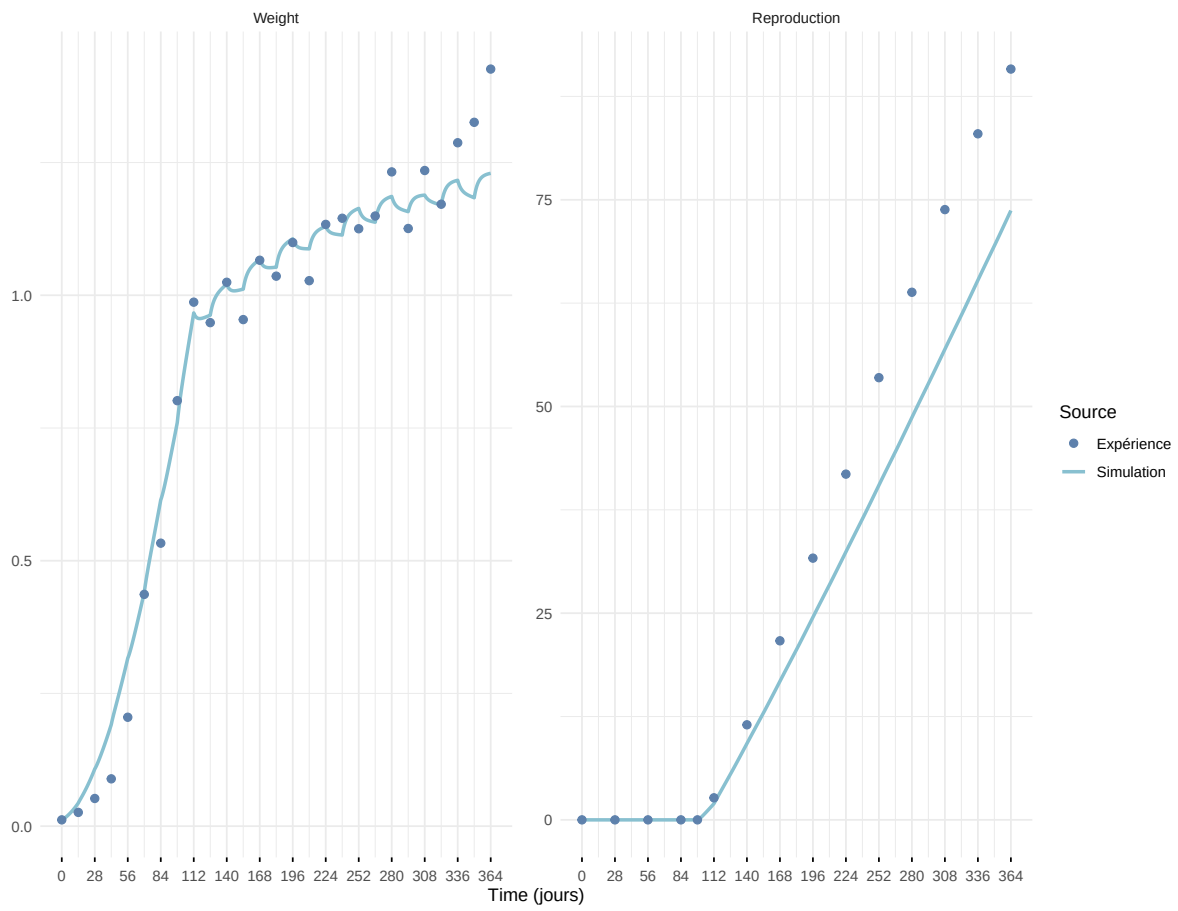


Figure 6: Simulations Weight and Reproduction.

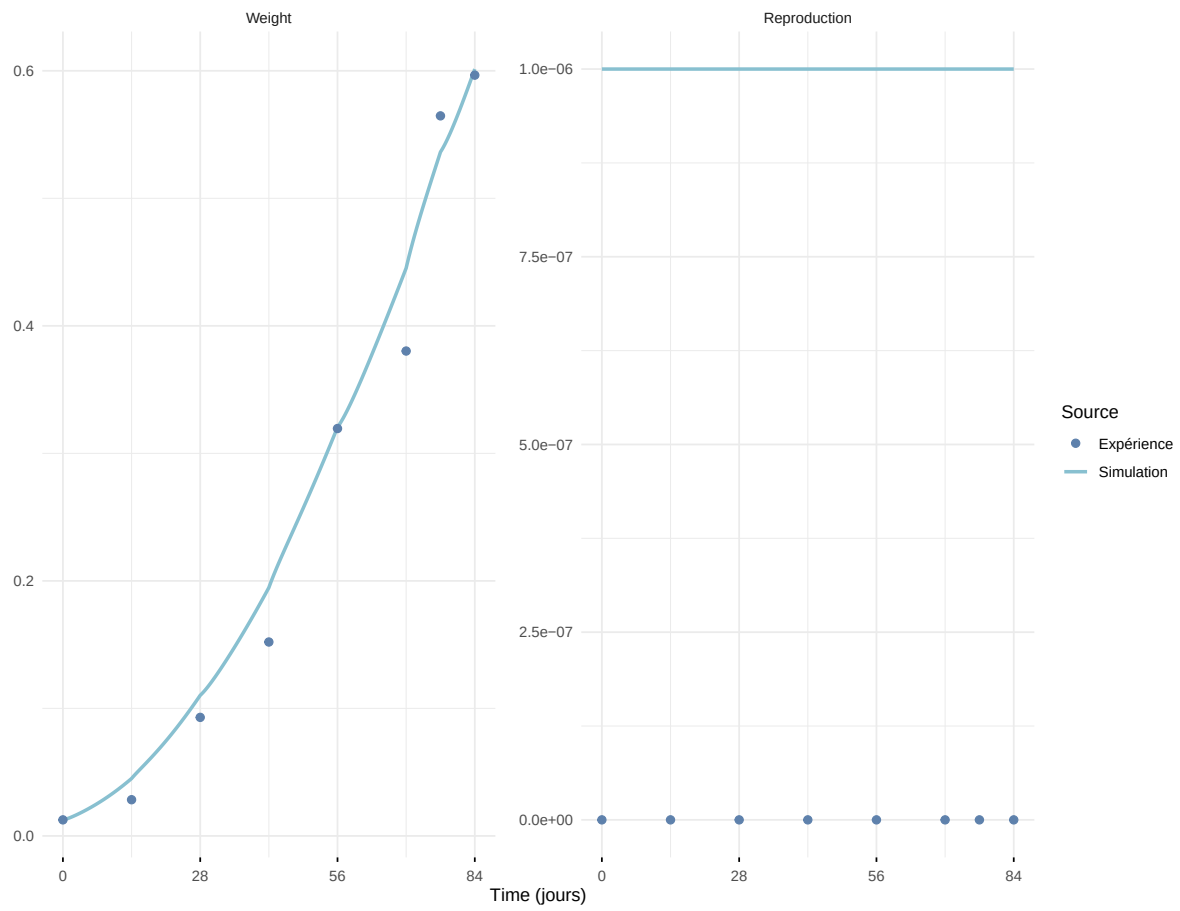
Expérience : Bart2019_XP1_1



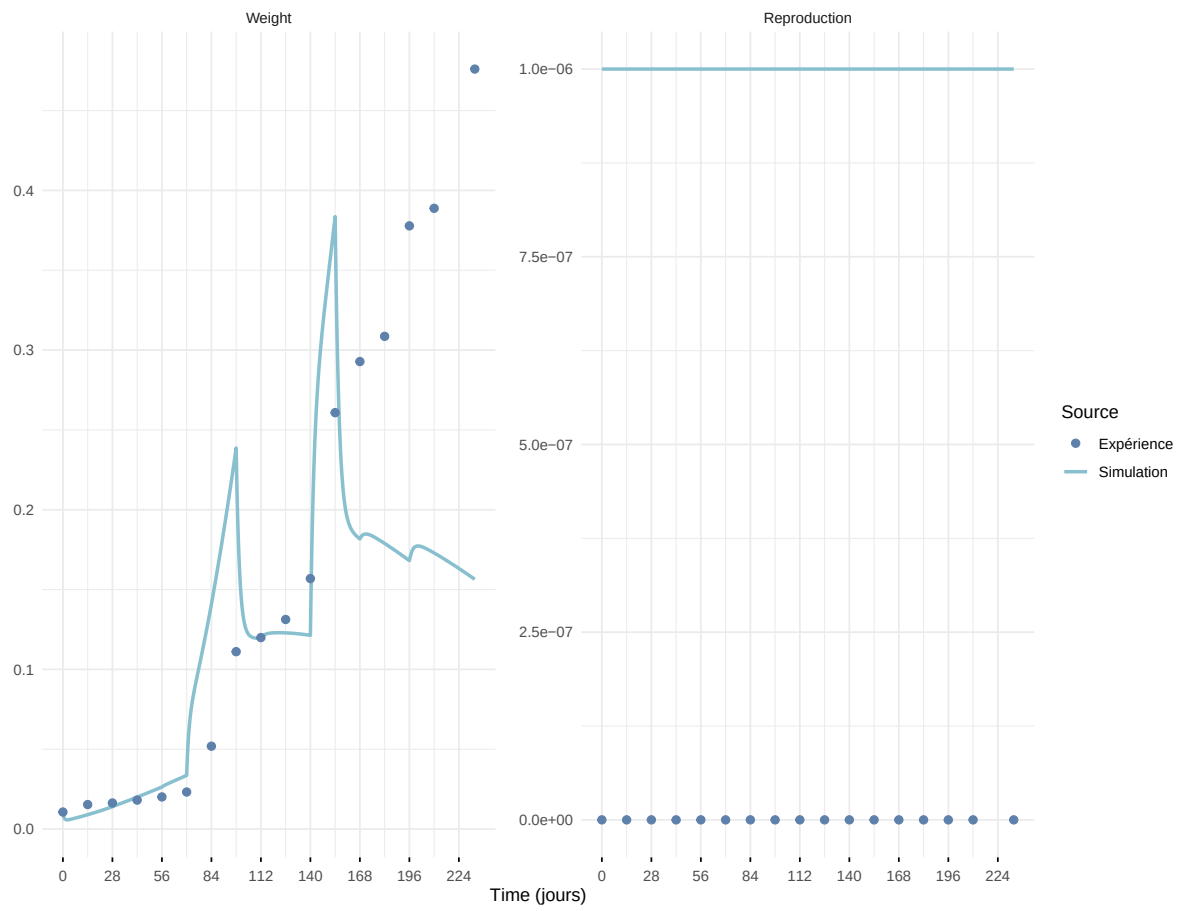
Expérience : Bart2019_XP1_2



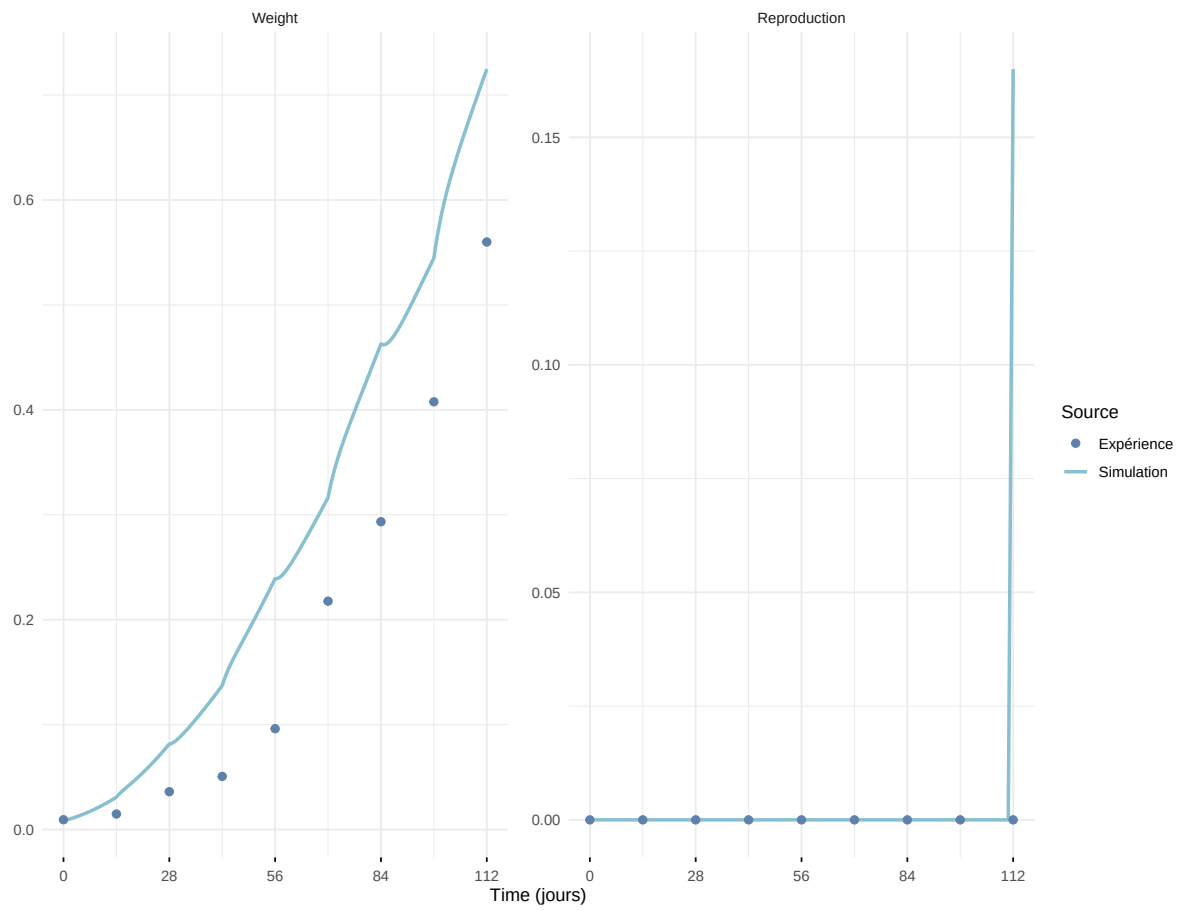
Expérience : Bart2019_XP2



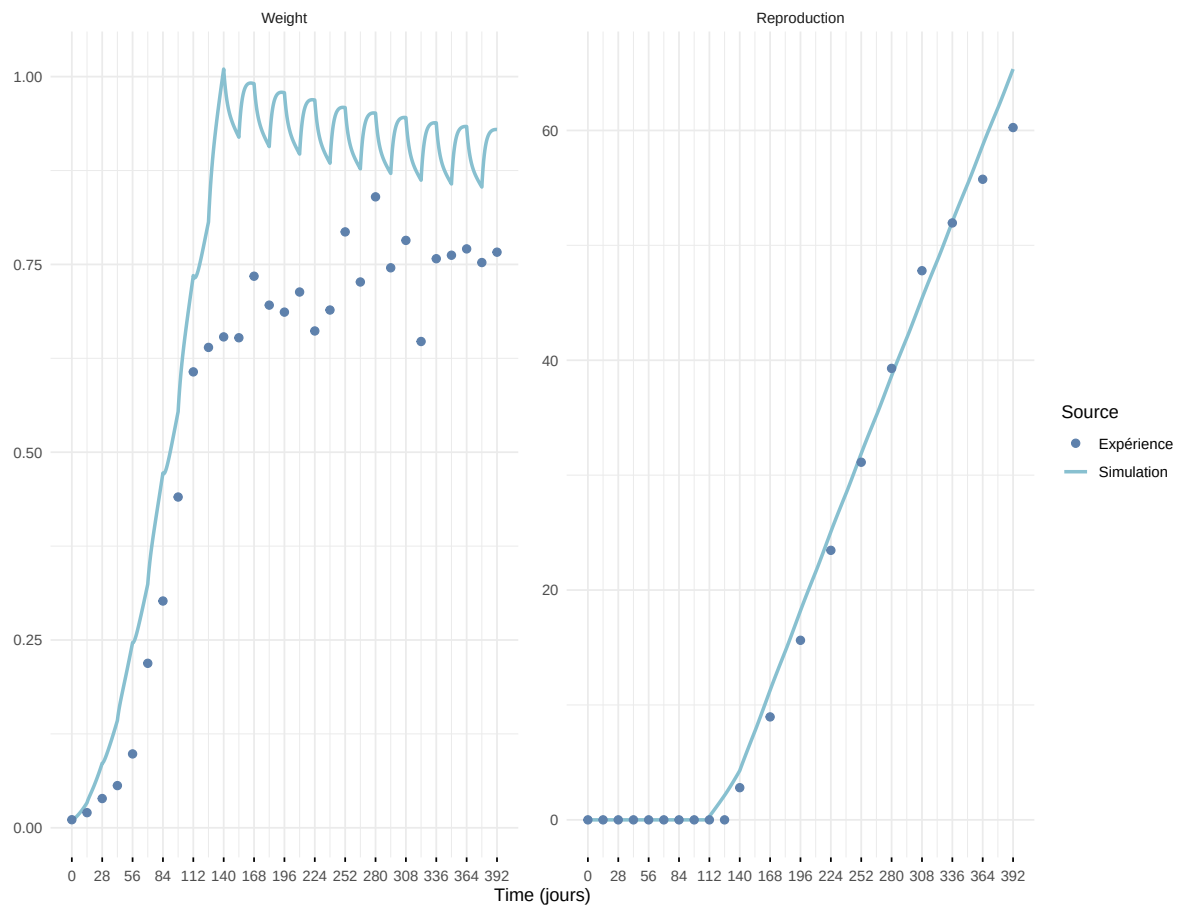
Expérience : Bart2019_XP3



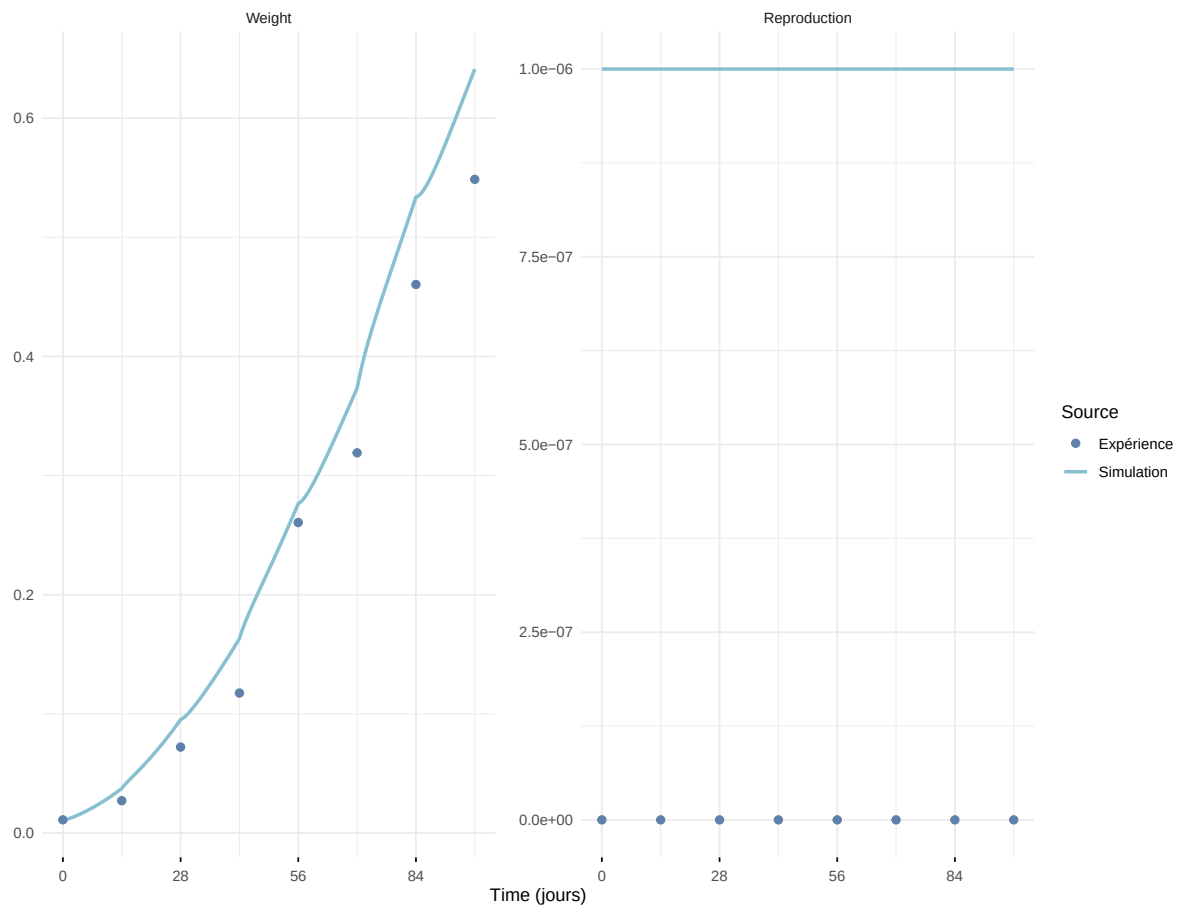
Expérience : Bart2019_XP4_1



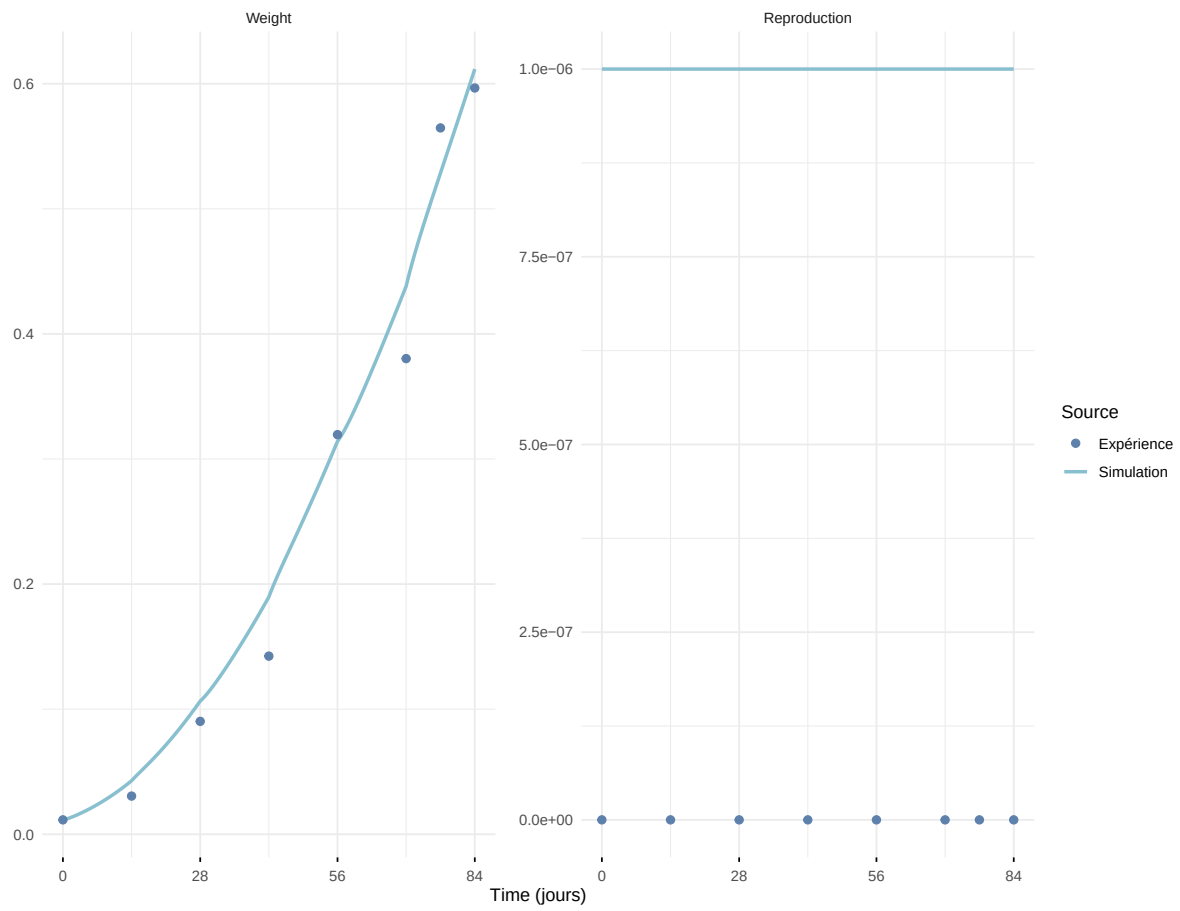
Expérience : Bart2019_XP4_2



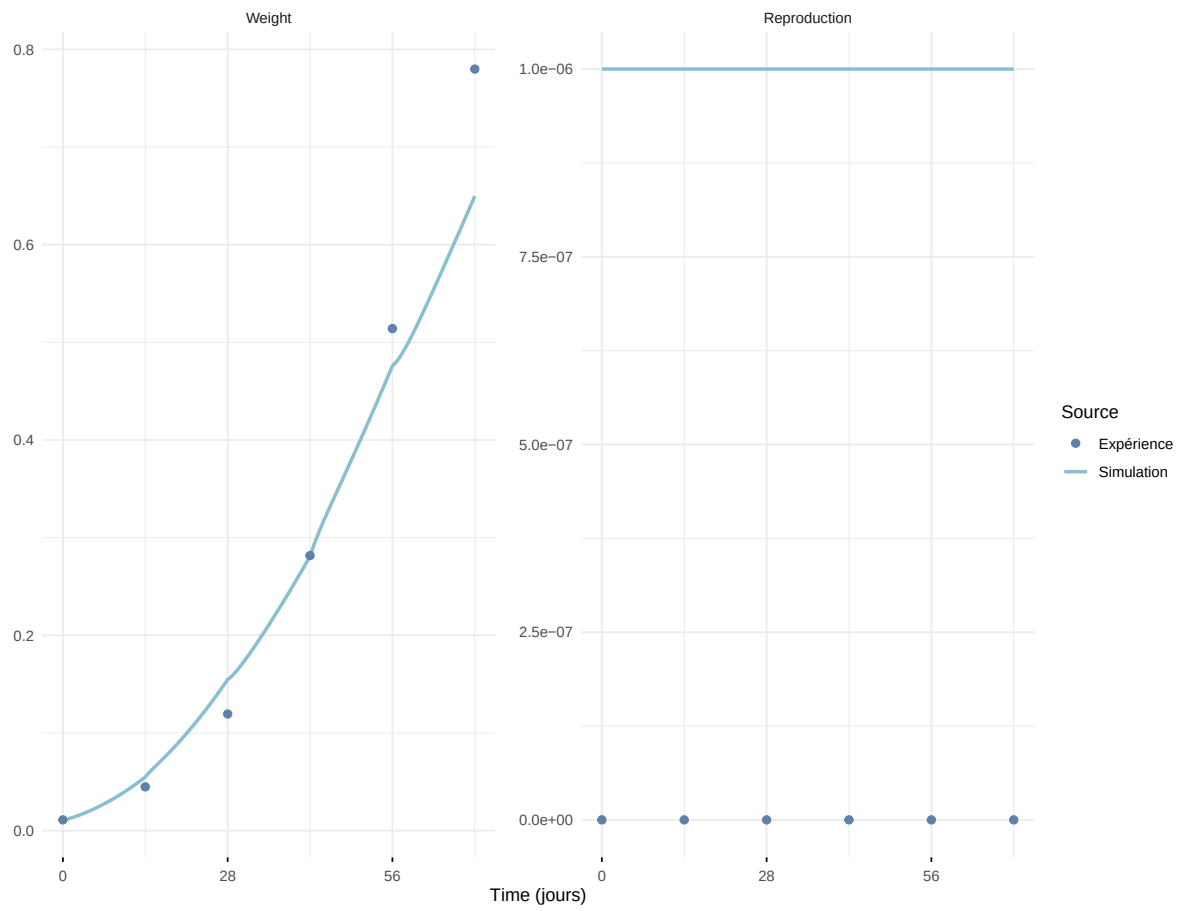
Expérience : Bart2019_XP5



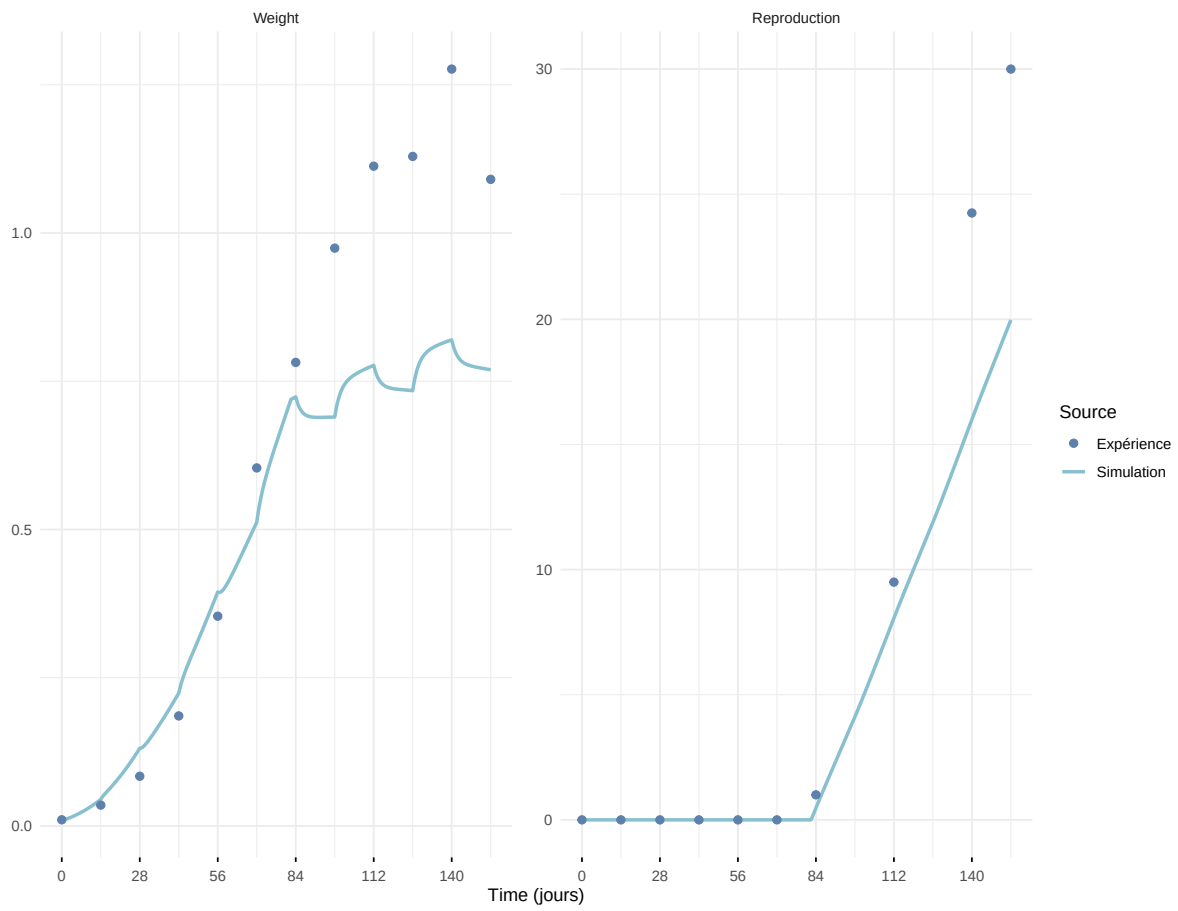
Expérience : Bart2020



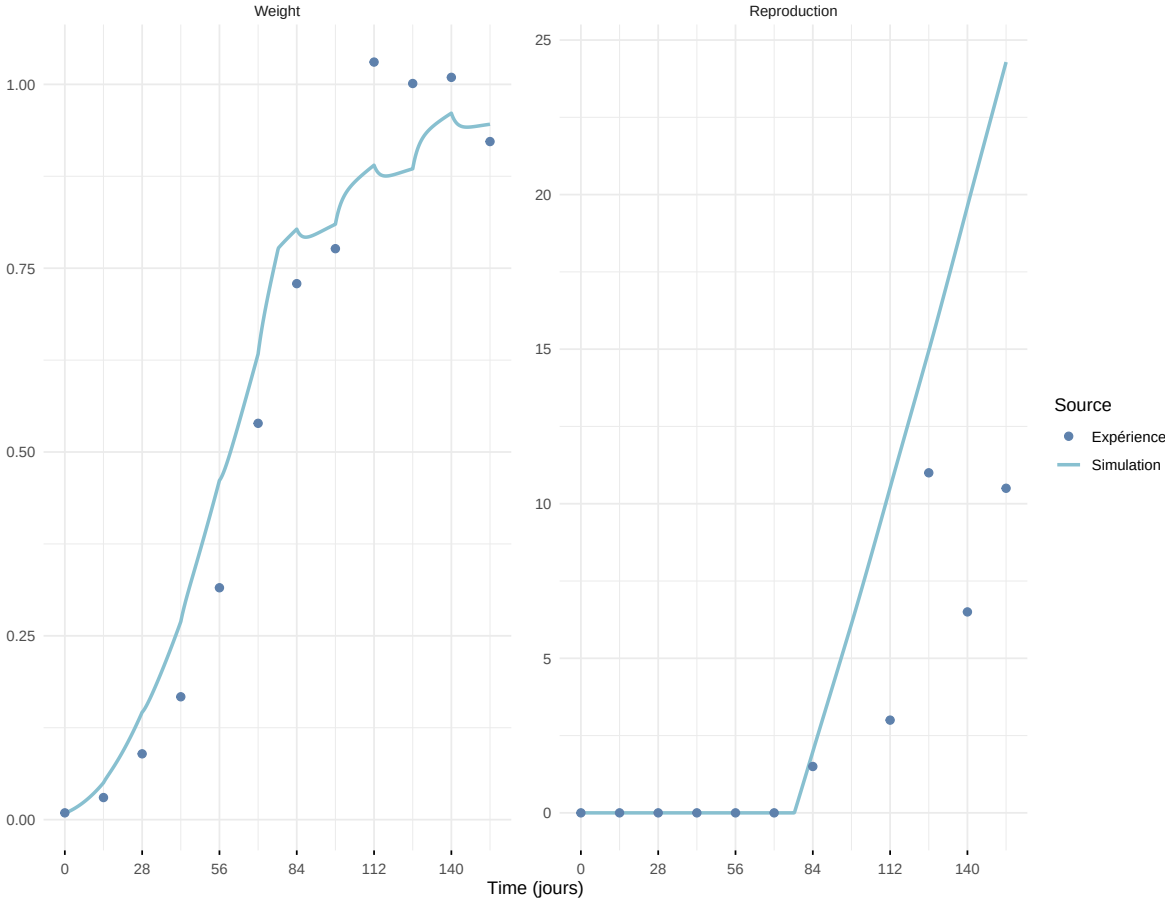
Expérience : Gollot2026_dens_D1



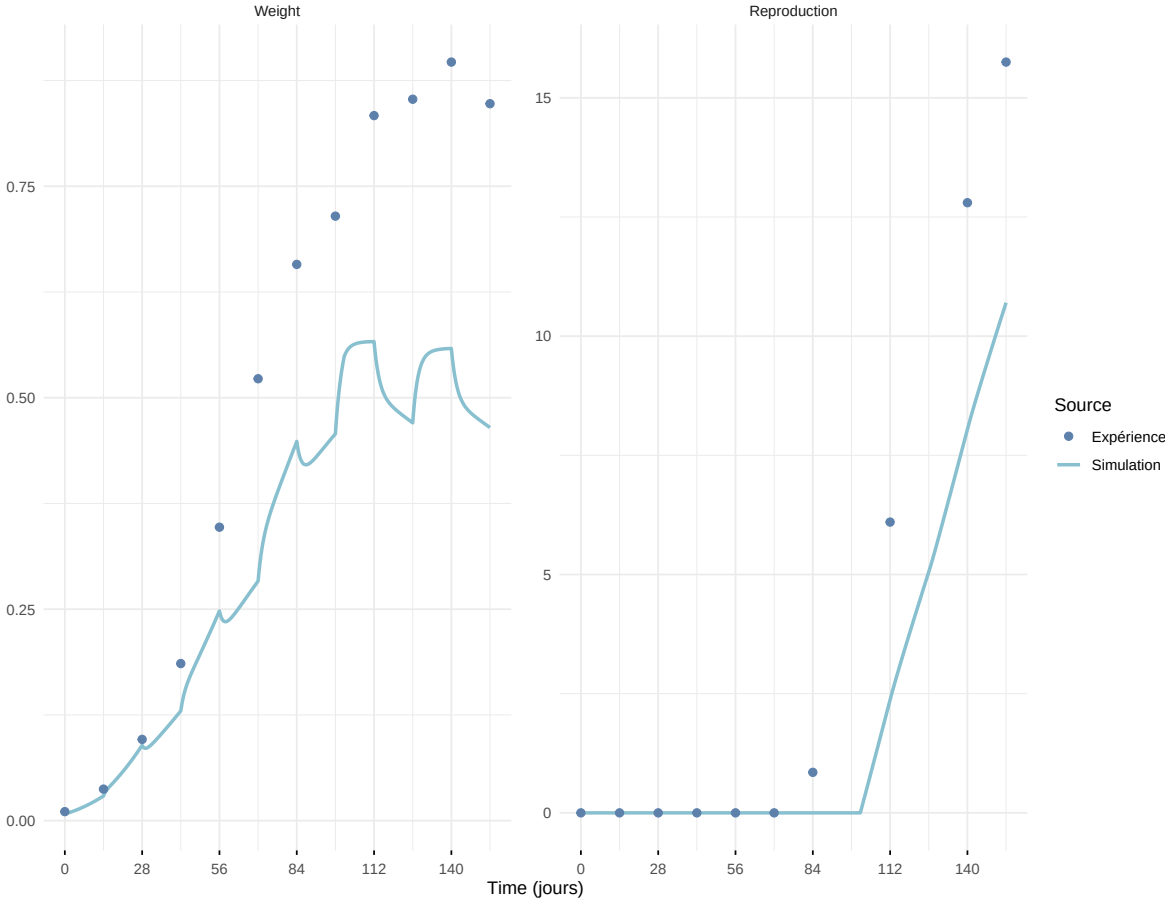
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

